Are-Perfomance-Rate-Given-Consistently–.R

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2026-01-08

```r
# Load the packages
library("readr")
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library("broom")
library("ggplot2")

# Import the data
hr_data <- read_csv("~/Desktop/Data/hr_data.csv" )
```

```
## Rows: 1470 Columns: 4

## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (3): department, job_level, gender
## dbl (1): employee_id
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
performance_data <- read_csv("~/Desktop/Data/performance_data.csv" )
```

```
## Rows: 1470 Columns: 2
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (2): employee_id, rating
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Examine the datasets
summary(hr_data)
```

```
##   employee_id      department          job_level            gender
##  Min.   :   1.0   Length:1470        Length:1470        Length:1470
##  1st Qu.: 491.2   Class :character   Class :character   Class :character
##  Median :1020.5   Mode  :character   Mode  :character   Mode  :character
##  Mean   :1024.9
##  3rd Qu.:1555.8
##  Max.   :2068.0
```

```r
summary(performance_data)
```

```
##   employee_id         rating
##  Min.   :   1.0   Min.   :1.00
##  1st Qu.: 491.2   1st Qu.:2.00
##  Median :1020.5   Median :3.00
##  Mean   :1024.9   Mean   :2.83
##  3rd Qu.:1555.8   3rd Qu.:4.00
##  Max.   :2068.0   Max.   :5.00
```

```r
# Join the two tables
joined_data <- left_join(hr_data, performance_data, by = "employee_id")

# Examine the result
summary(joined_data)
```

```
##   employee_id      department          job_level            gender
##  Min.   :   1.0   Length:1470        Length:1470        Length:1470
##  1st Qu.: 491.2   Class :character   Class :character   Class :character
##  Median :1020.5   Mode  :character   Mode  :character   Mode  :character
##  Mean   :1024.9
##  3rd Qu.:1555.8
##  Max.   :2068.0
##      rating
##  Min.   :1.00
##  1st Qu.:2.00
##  Median :3.00
##  Mean   :2.83
##  3rd Qu.:4.00
##  Max.   :5.00
```

```r
# Check whether the average performance rating differs by gender
joined_data %>%
  group_by(gender)%>%
  summarize(avg_rating = mean(rating ) )
```

```
## # A tibble: 2 x 2
##   gender avg_rating
##   <chr>       <dbl>
## 1 Female       2.75
## 2 Male         2.92
```

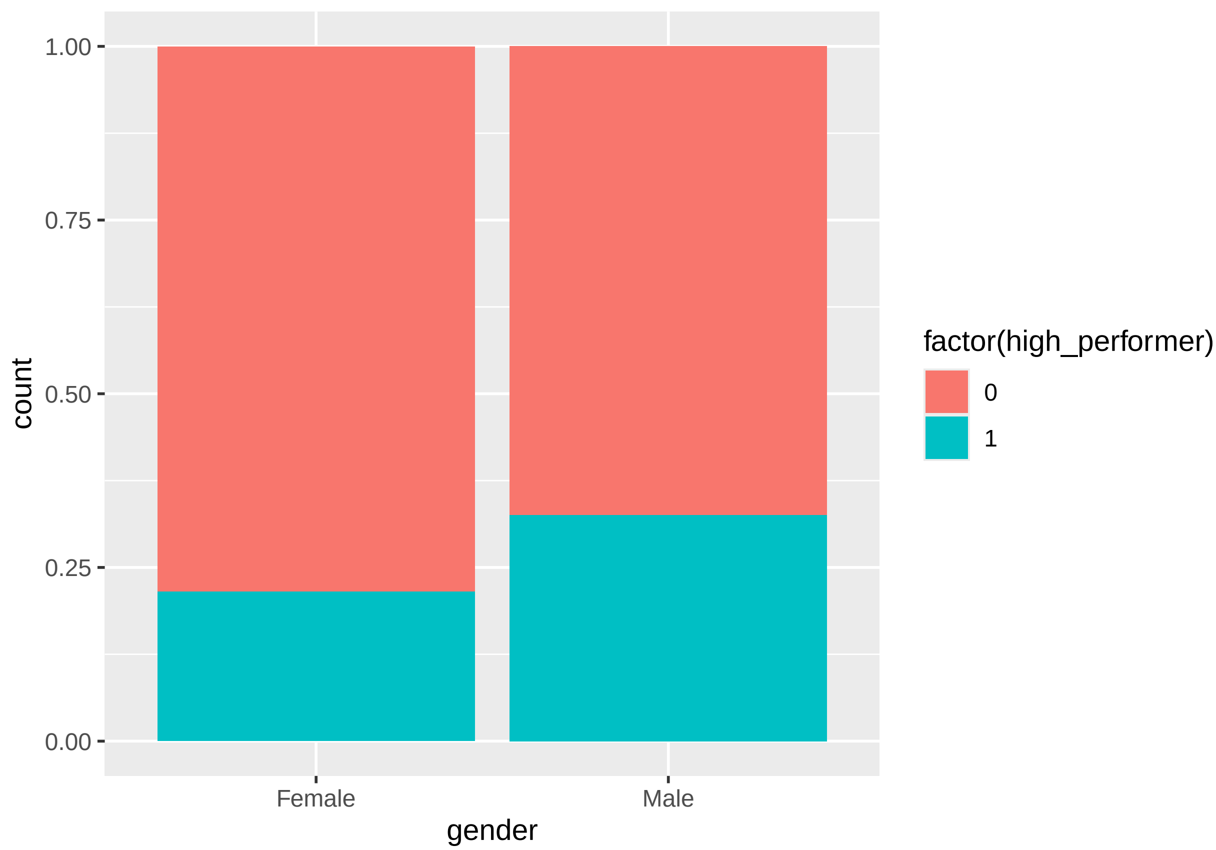
```r
# Add the high_performer column
performance <- joined_data %>%
  mutate(high_performer = ifelse(rating >= 4, 1, 0))

# Test whether one gender is more likely to be a high performer
chisq.test(performance$gender, performance$high_performer)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  performance$gender and performance$high_performer
## X-squared = 22.229, df = 1, p-value = 2.42e-06
```

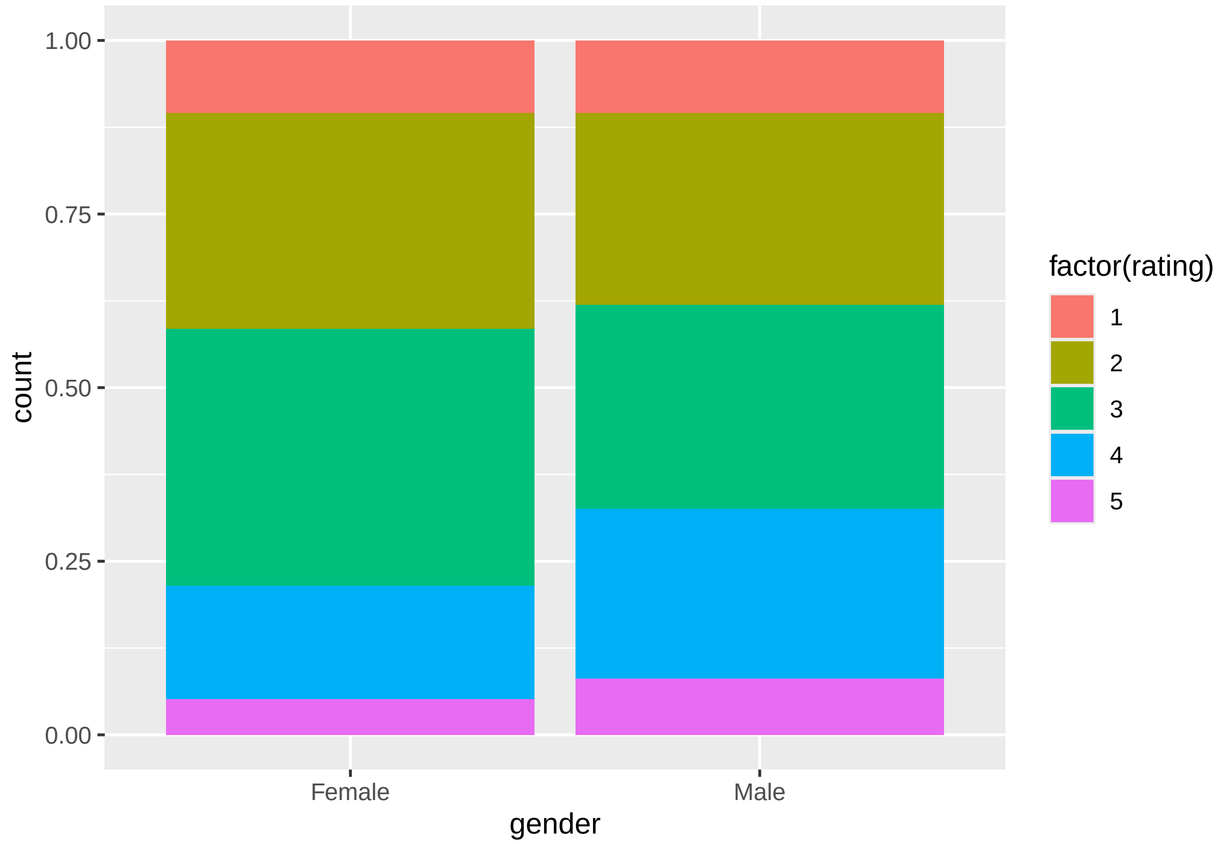
```r
# Do the same test, and tidy the output
chisq.test(performance$gender, performance$high_performer) %>%
  tidy()
```

```
## # A tibble: 1 x 4
##   statistic    p.value parameter method
##       <dbl>      <dbl>     <int> <chr>
## 1      22.2 0.00000242         1 Pearson's Chi-squared test with Yates' continu~
```

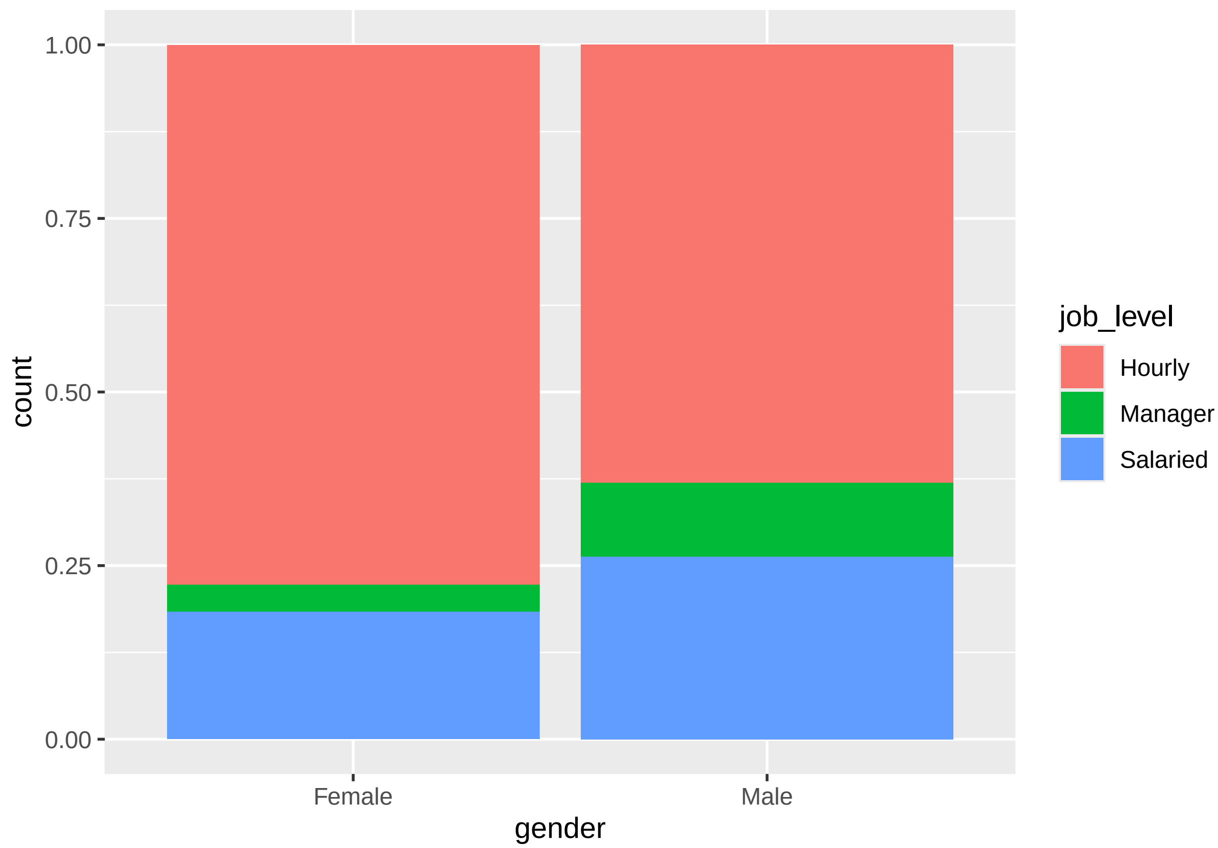
```r
# Visualize the distribution of high_performer by gender
ggplot(performance, aes(x=gender, fill = factor(high_performer))) + geom_bar(position = "fill")
```



```r
# Visualize the distribution of all ratings by gender
ggplot(performance, aes(x=gender, fill = factor(rating))) + geom_bar(position = "fill")
```



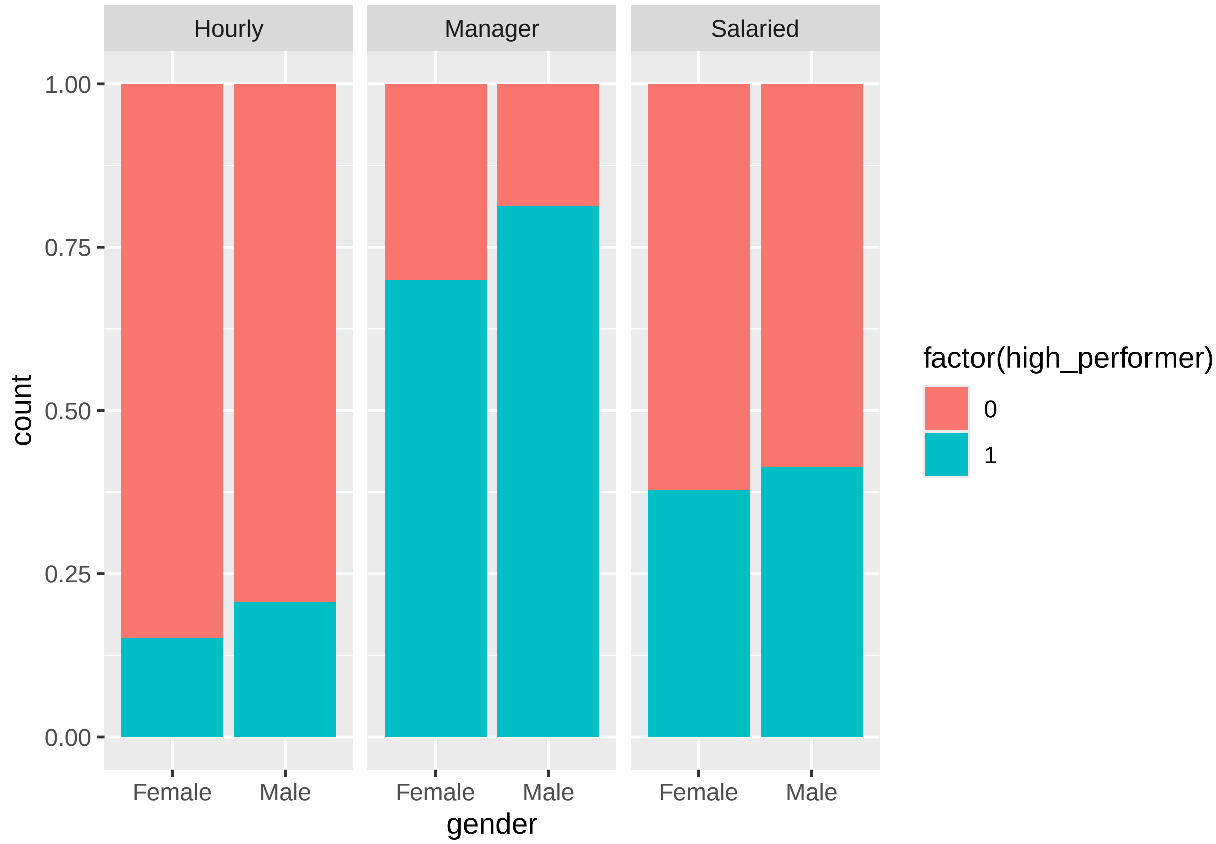

```r
# Visualize the distribution of job_level by gender
performance %>%
  ggplot(aes(x = gender, fill = job_level)) +
  geom_bar(position = "fill")
```




```
# Test whether men and women have different job level distributions
chisq.test(performance$gender, performance$job_level)
```

```
##
##  Pearson's Chi-squared test
##
## data:  performance$gender and performance$job_level
## X-squared = 44.506, df = 2, p-value = 2.166e-10
```

```
# Visualize the distribution of high_performer by gender, faceted by job level
performance %>%
  ggplot(aes(x = gender, fill = factor(high_performer))) +
  geom_bar(position = "fill") +
  facet_wrap(~ job_level)
```




```
# Run a simple logistic regression
logistic_simple <- glm(high_performer ~ gender, family = "binomial", data = performance)

# View the result with summary()
logistic_simple %>%
  summary()
```

```
##
## Call:
## glm(formula = high_performer ~ gender, family = "binomial", data = performance)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.29540    0.08813 -14.699  < 2e-16 ***
## genderMale   0.56596    0.11921   4.748 2.06e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 1709.0  on 1469  degrees of freedom
## Residual deviance: 1686.1  on 1468  degrees of freedom
## AIC: 1690.1
##
## Number of Fisher Scoring iterations: 4
```

```r
# View a tidy version of the result
logistic_simple %>%
  tidy()
```

```
## # A tibble: 2 x 5
##   term        estimate std.error statistic  p.value
##   <chr>          <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)   -1.30     0.0881    -14.7  6.58e-49
## 2 genderMale     0.566    0.119       4.75 2.06e- 6
```

```r
# Run a multiple logistic regression
logistic_multiple <- glm(high_performer ~ gender + job_level, family = "binomial", data = performance)

# View the result with tidy()
logistic_multiple %>%
  tidy()
```

```
## # A tibble: 4 x 5
##   term              estimate std.error statistic  p.value
##   <chr>                <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)         -1.69      0.103    -16.5  2.74e-61
## 2 genderMale           0.319     0.129      2.47 1.34e- 2
## 3 job_levelManager     2.74      0.251     10.9  1.01e-27
## 4 job_levelSalaried    1.10      0.141      7.82 5.17e-15
```

```r
# Undertake an initiative to ensure performance ratings are given fairly.
```